ESSENTIALLY FINITE TYPE CRITERION: SEPARATEDNESS PART

THE STACKS PROJECT AUTHORS

Editorial note. No omissions: the diagonal argument is worked; the sibling criteria are delivered cards cited as proved chapter lemmas.

Allowed standard prerequisites (editorial).

- DVR valuative separation. Limits, Lemma lemma-Noetherian-dvr-valuative-separation (same chapter)
- Ess-ft universally closed criterion. Limits, Lemma lemma-essentially-finite-type-criterion-universally-closed (same chapter)

Source reference. Original reference 00PD: sources/stacks-chow/tags.txt, source lines 927–927. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0GWT: sources/stacks-chow/tags.txt, source lines 21083–21083. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

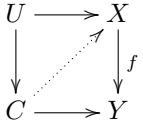
Source reference. Original lemma 0207: sources/stacks-chow/tags.txt, source lines 2624–2624. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01T7: sources/stacks-chow/tags.txt, source lines 2364–2364. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0GWV: sources/stacks-chow/tags.txt, source lines 21085–21085. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a Nagata scheme (and in particular locally Noetherian). Let $f : X \to Y$ be a morphism of schemes locally of finite type over S. The following are equivalent*

(1) *f separated,*
(2) *for any commutative diagram*



of schemes over S such that
(a) *C is a normal integral scheme of finite type over S,*
(b) *$U = C \setminus \{c\}$ for some closed point $c \in C$,*

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(c) $A = \mathcal{O}_{C,c}$ *has dimension* 1[1]
then in the commutative diagram

$$\begin{array}{ccc} \mathrm{Spec}(K) & \longrightarrow & X \\ \downarrow & \nearrow & \downarrow f \\ \mathrm{Spec}(A) & \longrightarrow & Y \end{array}$$

where $K = Frac(A)$ *there exists at most one dotted arrow*[2] *making the diagram commute.*

Proof. By Lemma 0207 we see that (1) implies (2). Assume (2). In order to show that f is separated, we have to show that $\Delta : X \to X \times_Y X$ is closed. By Morphisms, Lemma 01T7 the morphism Δ is quasi-compact. By Lemma 0GWV it suffices to show: for any commutative diagram

$$\begin{array}{ccc} U & \longrightarrow & X \\ \downarrow & \overset{(a_1,a_2)}{\nearrow} & \downarrow \Delta \\ C & \longrightarrow & X \times_Y X \end{array}$$

of schemes over S such that

(1) C is a normal integral scheme of finite type over S,
(2) $U = C \setminus \{c\}$ for some closed point $c \in C$,
(3) $A = \mathcal{O}_{C,c}$ has dimension 1.

then in the commutative diagram

$$\begin{array}{ccc} \mathrm{Spec}(K) & \longrightarrow & X \\ \downarrow & \nearrow & \downarrow \Delta \\ \mathrm{Spec}(A) & \longrightarrow & X \times_Y X \end{array}$$

where $K = \mathrm{Frac}(A)$ there exists some dotted arrow making the diagram commute. By Lemma 0GWT the existence of the dotted arrow in the second diagram is equivalent to the existence of the dotted arrow in the first diagram. Moreover, the existence there is the same as asking $a_1 = a_2$. However $a_1|_U = a_2|_U$, so by the uniqueness assumption (2) we see that this is true and the proof is complete. □

[1] It follows that A is a discrete valuation ring, see Algebra, Lemma 00PD. Moreover, c maps to a finite type point $s \in S$ and A is essentially of finite type over $\mathcal{O}_{S,s}$.

[2] By Lemma 0GWT this is equivalent to asking there to be at most one dotted arrow making the first commutative diagram commute.